BTS-IV-04.19-0360 Reg. No. | | | | | | | | |

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B.Tech. Degree IV Semester Examination April 2019

CE 1405 (A/B) FLUID MECHANICS II

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Briefly explain the classification of flow in open channels.

(b) A flow of water of 100 litres per second flows down in a rectangular flume of width 600 mm and having adjustable bottom slope. If Chezy's constant C is 56. Find the bottom slope necessary for uniform flow with a depth of flow of 300 mm. Also find the conveyance K of the flume.

(c) Explain the terms: (i) Specific energy of a flowing liquid (ii) Critical depth (iii) Critical velocity as applied to non-uniform flow.

(d) Explain the term hydraulic jump and derive an expression for it.

(e) How are hydraulic turbines classified?

(f) What is meant by governing of turbines?

(g) Define the terms: suction head, delivery head, static head and manometric head as applied to centrifugal pump.

(h) Define slip, percentage slip and negative slip of a reciprocating pump.

PART B

(4 × 15 = 60)

II. Prove that for the trapezoidal channel of most economical section (i) Half of top width = Length of one of the slopping sides (ii) Hydraulic mean depth = $\frac{1}{2}$ Depth of flow (iii) Three sides of the trapezoidal section will be tangential to the semi-circle described on the water line. (15)

OR

III. Prove that for the channel of circular section, (i) d = 0.81 D for maximum velocity (ii) d = 0.95 D for maximum discharge (iii) m = 0.3 D for maximum velocity, where D = diameter of circular channel, d= depth of flow and m = hydraulic mean depth. (15)

IV. (a) What is the essential difference between GVF and RVF? Illustrate with neatly drawn sketches. (7)

(b) Derive the differential equation for steady gradually varied flow in open channels and list all assumptions. (8)

OR

V. Determine the length of the back water curve caused by an affux of 2 m in a rectangular channel of width 40 m and depth 2.5 m. The slope of the bed is given as 1 in 11000. Take Manning's N = 0.3. (15)

(P.T.O.)

VI. A jet of water having a velocity of 15 m/s strikes a curved vane which is moving with a velocity of 5 m/s. The vane is symmetrical and it is so shaped that the jet is deflected through $120°$. Find the angle of the jet at inlet of the vane so that there is no shock. What is the absolute velocity of the jet at outlet in magnitude and direction and the work done per unit weight of water? Assume the vane to be smooth. (15)

OR

VII. What is a draft tube? Why is it used in reaction turbine? What are the advantages of draft tube? Describe with neat sketches, different types of draft tubes. (15)

VIII. With a neat sketch, explain the principle, parts and working of a centrifugal pump. Also obtain an expression for the work done by the impeller of a centrifugal pump on water per second per unit weight of water. (15)

OR

IX. A single acting reciprocating pump has a plunger of diameter 200 mm and stroke of 300 mm. If the speed of the pump is 60 rpm and delivers 15.5 litres per second of water aginast a suction head of 5 m and a delivery head of 20 m, find the theoretical discharge of the pump, co-efficient of discharge, the slip, the percentage slip of the pump and power required to drive the pump. (15)

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